

Q4. Docker Network:-

- Communication between two containers through a Docker network.
- Communication using container names and container IP addresses.

1. Create a Docker network

```
srishti2902@LAPTOP-6LVIDNI x + v
ayushi2323@LAPTOP-6LVIDNHT:~# docker network create mynetwork
ef17hab37cebbface4d04bbe978ec896416512999a475ee2fb1afe29fd5f9cf9
ayushi2323@LAPTOP-6LVIDNHT:~# docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
135790a3248d        bridge             bridge              local
c3bbb4b3f17d        host               host                local
ef17bab37ceb        mynetwork          bridge              local
11e0b59157cb        none              null                local
```

2. Create the first container

```
ayushi2323@LAPTOP-6LVIDNHT:~# docker run -dit --name container1 --network mynetwork ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
f3db1cd94078: Pull complete
06e9d71331fb: Pull complete
f79afc4bbefe: Download complete
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Downloaded newer image for ubuntu:latest
4a46e330eac6ca107de22f7dfd9a90364ccb468a7c0db9754d04f728699a8d5
```

3. Create the second container

```
ayushi2323@LAPTOP-6LVIDNHT:~$ docker run -dit --name container2 --network mynetwork ubuntu
docker: Error response from daemon: Conflict. The container name "/container2" is already in use by container "ayushi2323/container1". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
ayushi2323@LAPTOP-6LVIDNHT:~$ docker run -dit --name container2 --network mynetwork ubuntu
5e91089998f1ee1eb8a32974ab1df3cd76e0d5f43223197aa055254971ca505
ayushi2323@LAPTOP-6LVIDNHT:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
5e91089998f3   ubuntu   "/bin/bash"             9 seconds ago Up 9 seconds          container2
4a46e330eac6   ubuntu   "/bin/bash"             36 seconds ago Up 33 seconds          container1
ce1b7cc14510   nginx    "/docker-entrypoint..." 20 minutes ago Up 20 minutes    80/tcp       volume-test-2
```

4. Install ping in both containers

```

ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec container1 apt-get update
docker exec container1 apt-get install -y iptutils-ping
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [578 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [189 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [212 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [11.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [743 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [14.6 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [585 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [338 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3306 B]
Fetched 26.0 MB in 9s (2820 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
Solving dependencies...
The following additional packages will be installed:
  libcap2 libidn2-0 libunistring5 linux-sysctl-defaults
The following NEW packages will be installed:
  iptutils-ping libcap2 libidn2-0 libunistring5 linux-sysctl-defaults
0 upgraded, 5 newly installed, 0 to remove and 24 not upgraded.
Need to get 761 kB of archives.
After this operation, 2677 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcap2 amd64 1:2.75-1ubuntu2 [29.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunistring5 amd64 1.3-2build1 [618 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libidn2-0 amd64 2.3.8-4build1 [67.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 iptutils-ping amd64 3:20250605-1ubuntu1 [47.1 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 linux-sysctl-defaults all 4.15ubuntu5 [6288 B]
debconf: unable to initialize frontend: Dialog
debconf: (TERM is not set, so the dialog frontend is not usable.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline

```

```

ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec container2 apt-get update
docker exec container2 apt-get install -y iptutils-ping
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [11.2 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [212 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [189 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [578 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]

```

1. Communication using container name

```

srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container1 ping -c 4 container2
PING container2 (172.18.0.3) 56(84) bytes of data:
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=1 ttl=64 time=3.76 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=2 ttl=64 time=0.312 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=3 ttl=64 time=0.188 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=4 ttl=64 time=0.123 ms

--- container2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.123/1.094/3.755/1.537 ms

```

2. Find the IP address of container2

```

srishti2902@LAPTOP-6LVIDNHT:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' container2
172.18.0.3

```

3. Communication using container IP address

```

srishti2902@LAPTOP-6LVIDNHT:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' container2
172.18.0.3
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container1 ping -c 4 172.18.0.3
PING 172.18.0.3 (172.18.0.3) 56(84) bytes of data:
64 bytes from 172.18.0.3: icmp_seq=1 ttl=64 time=0.155 ms
64 bytes from 172.18.0.3: icmp_seq=2 ttl=64 time=0.098 ms
64 bytes from 172.18.0.3: icmp_seq=3 ttl=64 time=0.096 ms
64 bytes from 172.18.0.3: icmp_seq=4 ttl=64 time=0.120 ms

--- 172.18.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3074ms
rtt min/avg/max/mdev = 0.096/0.117/0.155/0.023 ms

```

4. Check the network details

```

ayushi2323@LAPTOP-6LVIDNHT:~# docker network inspect mynetwork
[
  {
    "Name": "mynetwork",
    "Id": "ef17bab77cebbface4d84bbe57bee89661641375ea475ee2fblafe29fd5f9cf9",
    "Created": "2026-09-08T14:06:48.067539661Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": true,
    "EnableIPv4": false,
    "IPAM": {
      "Subnet": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": [],
    "Containers": {
      "1a40e636aeaf6ca107de2277dfd9a98364acb668a7c8db4754d84f738699a8d5": {
        "Name": "container1",
        "EndpointID": "10a888331e3503fdb828a73fc8e54876bd6716881cb0fd24be194d9533eed9b",
        "MacAddress": "e6:1b:78:85:fe:2a",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      },
      "5a91289998f5ee1eb9a33374ab1df2cd76e8d5f43223197ea8855254371ca585": {
        "Name": "container2",

```

1. Test communication in the opposite direction

```

ayushi2323@LAPTOP-6LVIDNHT:~# docker inspect -f '{{range .NetworkSettings.Networks}}
172.18.0.2
ayushi2323@LAPTOP-6LVIDNHT:~# docker exec container2 ping -c 4 container1
PING container1 (172.18.0.2) 64(80) bytes of data:
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=1 ttl=64 time=0.243 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=2 ttl=64 time=0.485 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=3 ttl=64 time=0.144 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=4 ttl=64 time=0.086 ms

--- container1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3046ms
rtt min/avg/max/mdev = 0.086/0.239/0.485/0.152 ms
ayushi2323@LAPTOP-6LVIDNHT:~# docker exec container2 ping -c 4 <container1-IP>
-bash: syntax error near unexpected token 'newline'
ayushi2323@LAPTOP-6LVIDNHT:~# docker exec container2 ping -c 4 <container1-172.
-bash: syntax error near unexpected token 'newline'
ayushi2323@LAPTOP-6LVIDNHT:~# docker exec container2 ping -c 4 172.18.0.2
PING 172.18.0.2 (172.18.0.2) 64(80) bytes of data:
64 bytes from 172.18.0.2: icmp_seq=1 ttl=64 time=0.215 ms
64 bytes from 172.18.0.2: icmp_seq=2 ttl=64 time=0.088 ms
64 bytes from 172.18.0.2: icmp_seq=3 ttl=64 time=0.100 ms
64 bytes from 172.18.0.2: icmp_seq=4 ttl=64 time=0.056 ms

--- 172.18.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3046ms
rtt min/avg/max/mdev = 0.086/0.121/0.212/0.052 ms

```

2. Final cleanup

```

ayushi2323@LAPTOP-6LVIDNHT:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}
172.18.0.2
ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 container1
PING container1 (172.18.0.2) 56(84) bytes of data.
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=1 ttl=64 time=0.243 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=2 ttl=64 time=0.485 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=3 ttl=64 time=0.144 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=4 ttl=64 time=0.086 ms

--- container1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3046ms
rtt min/avg/max/mdev = 0.086/0.239/0.485/0.152 ms
ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 <container1-IP>
-bash: syntax error near unexpected token `newline'
ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 <container1-172.18.0.2>
-bash: syntax error near unexpected token `newline'
ayushi2323@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 172.18.0.2
PING 172.18.0.2 (172.18.0.2) 56(84) bytes of data.
64 bytes from 172.18.0.2: icmp_seq=1 ttl=64 time=0.212 ms
64 bytes from 172.18.0.2: icmp_seq=2 ttl=64 time=0.088 ms
64 bytes from 172.18.0.2: icmp_seq=3 ttl=64 time=0.100 ms
64 bytes from 172.18.0.2: icmp_seq=4 ttl=64 time=0.086 ms

--- 172.18.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.086/0.121/0.212/0.052 ms

```

1. Final cleanup

```

ayushi2323@LAPTOP-6LVIDNHT:~$ docker stop container1 container2
container1
container2
ayushi2323@LAPTOP-6LVIDNHT:~$ docker rm container1 container2
container1
container2
ayushi2323@LAPTOP-6LVIDNHT:~$ docker network rm mynetwork
mynetwork
ayushi2323@LAPTOP-6LVIDNHT:~$

```